
ARCA: Adapter-Residual Credit Assignment When Token Signals Collapse

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Abstract

1 Token-level credit assignment for language-model reinforcement learning is usually
2 formulated as if the policy were fully trainable, while practical LLM-RL pipelines
3 overwhelmingly rely on parameter-efficient fine-tuning, especially LoRA. We ar-
4 gue that this separation hides a structural failure mode. Under LoRA, the policy is
5 restricted to a low-rank neighbourhood of the reference model, so the per-token
6 output-distribution differences used by common intrinsic credit signals—surprisal,
7 entropy reduction, and policy divergence—can be driven toward degenerate nor-
8 malized weights. We formalize this collapse and propose measuring it directly
9 with concentration diagnostics such as weight Gini and effective-token ratio. We
10 then introduce *Adapter-Residual Credit Assignment* (ARCA), a lightweight alter-
11 native that derives token salience from the adapter’s own hidden-state residual,
12 $\|h_t^{\text{adapted}} - h_t^{\text{base}}\|_2$. ARCA asks where the adapter actually changes the model,
13 rather than where the output distribution appears uncertain or shifted, and requires
14 no reward model, value head, or tree construction. In a compact MATH/Qwen3-
15 1.7B GRPO sweep, ARCA exhibits the predicted non-collapsed middle-regime
16 credit distribution under matched rollout budgets and remains competitive with
17 rank-matched baselines.

18 1 Introduction

19 Reinforcement learning has become a central component of large language model (LLM) post-training,
20 particularly for alignment and for reasoning-oriented tasks with verifiable rewards. A persistent
21 challenge across these settings is *credit assignment*: trajectories are long, rewards are sparse and
22 outcome-level, and it is not obvious how a single scalar signal at the end of a generation should be
23 distributed across hundreds or thousands of token decisions. Recent work has responded with a rapidly
24 growing toolbox of token-level credit-assignment methods, including entropy-aware modulation,
25 process reward models, tree-based prefix values, and reward redistribution [7, 16, 32, 34, 9, 41, 14].

26 In parallel, essentially all open-source and academic LLM-RL training uses parameter-efficient
27 fine-tuning, most commonly LoRA [11]. This is an empirical reality driven by memory and compute
28 constraints: every widely used RL framework, from TRL to verl, uses LoRA by default, and recent
29 results show that LoRA + RL can be dramatically more sample- and parameter-efficient than full fine-
30 tuning [33]. Yet the two research threads are developed in near-total isolation. Papers on token-level
31 credit assignment specify methods without reference to the adaptation strategy, and PEFT is treated
32 as an orthogonal implementation detail.

33 In this work we argue that this separation is a mistake, and that the *interaction* between PEFT
34 and credit assignment is itself a first-class scientific object. The core issue is geometric. Intrinsic
35 token-level weighting schemes derive per-token salience from quantities such as surprisal, entropy
36 reduction, or divergence between the policy and a reference model. Under LoRA, however, the policy

is constrained to a small low-rank neighbourhood of the reference, and the per-token differences that these signals measure can lose the variation that made them useful. We formalize this as a collapse of the salience distribution toward degenerate token credit, characterize it with Gini coefficient and effective token count, and show that the same mechanism applies across multiple output-distribution-based weighting rules. This gives a mechanistic explanation for the empirically observed failure of full-token GRPO + LoRA [15]: the weighting signal itself is degraded before training ever begins.

Rather than trying to recover per-token structure from signals that LoRA intrinsically flattens, we measure salience directly from the adapter’s own contribution to the forward pass. Adapter-Residual Credit Assignment (ARCA) sets the salience at position t equal to the norm of the adapter residual, $\|h_t^{\text{adapted}} - h_t^{\text{base}}\|_2$, computed via a forward pass with the adapter disabled. This signal is positive whenever the adapter is active, non-uniform across positions because adapter input activations are non-uniform, and requires no additional networks, reward models, or tree construction. It reuses the same adapter-disabled forward pass already needed for KL regularization and divergence weighting.

Concretely, this paper makes three contributions:

1. We identify and formalize a PEFT-specific failure mode of token credit assignment: under LoRA, output-distribution salience can lose the per-token variation that intrinsic weighting methods rely on, even when the underlying RL objective is unchanged.
2. We introduce ARCA, a lightweight adapter-residual credit assignment rule whose normalized token weights remain non-degenerate whenever the adapter has position-varying hidden-state impact.
3. We validate the mechanism with concentration diagnostics and a matched MATH/Qwen3-1.7B sweep, separating downstream performance from the more basic question of whether a proposed token-credit signal survives the adaptation regime actually used in LLM-RL.

The remainder of the paper develops these contributions. Section 2 positions the work relative to PEFT for language RL and token-level credit assignment. Section 3 presents the weighting schemes, explains why output-distribution signals collapse under LoRA, and defines ARCA. Section 4 reports the diagnostic and performance comparison on MATH with Qwen3-1.7B under the seven-run submission-time sweep. Extended related work and theoretical interpretation appear in Appendices A and C.

2 Related Work

2.1 PEFT and Low-Rank Adaptation in Language RL

Parameter-efficient fine-tuning, and in particular LoRA [11], has become the default adaptation strategy for essentially all open-source LLM-RL work. Tina demonstrated that tiny LoRA adapters on a 1.5B base model suffice to reach DeepSeek-R1-class reasoning behaviour when combined with GRPO, at a tiny fraction of full fine-tuning cost [33]. A broader systematic evaluation of PEFT methods under RLVR [39] benchmarks over a dozen PEFT variants (including DoRA, AdaLoRA, MiSS, PiSSA, MiLoRA, VeRA and Rank-1) on DeepSeek-R1-Distill models and finds that standard LoRA is not optimal—structural variants such as DoRA, AdaLoRA and MiSS consistently outperform it, and SVD-initialised variants (PiSSA, MiLoRA) suffer from *spectral collapse*, a finding that is broadly compatible with our own signal-collapse analysis. Token-Efficient RL introduces critic-free LoRA-compatible variants of GRPO (S-GRPO and T-SPMO) that focus training on a subset of tokens, reporting large gains on small models where full-token GRPO + LoRA trains unstably [15]; we now provide a mechanistic explanation for that instability via signal collapse. *LoRA as an Implicit KL Regularizer* analyses how LoRA restricts the policy to a rank-constrained neighbourhood of the reference, deriving an explicit rank-dependent upper bound on the KL divergence between policy and reference throughout training [2]. Our Section 3.4 builds directly on this observation: an implicit KL bound is the same mechanism that forces per-token log-probability differentials to be small and approximately uniform, which is the formal content of our collapse result.

On the analysis side, *Narrow Finetuning Traces* shows that narrow finetuning leaves clearly readable traces in the activation differences between base and finetuned hidden states, and that the finetuning domain can be recovered from those differences alone using simple diffing tools such as patchscopes and activation steering [21]. This is directly relevant to ARCA because the adapter residual $h_t^{\text{adapted}} -$

89 h_t^{base} is a per-position activation difference of exactly the kind they study; their results supply
 90 empirical evidence that this difference carries meaningful, semantically non-trivial content—the
 91 property our method exploits. Concurrent work on TopLoRA studies how to concentrate LoRA
 92 capacity on a small number of high-impact tokens via token-wise input-output projections [17]. This
 93 can be viewed as a complementary response to the same underlying problem: TopLoRA modifies
 94 the adapter itself so that its capacity is allocated more selectively, while ARCA leaves the adapter
 95 unchanged but reweights the policy gradient so that token updates reflect where the adapter actually
 96 had non-trivial influence. To our knowledge, no prior work has characterized the interaction between
 97 LoRA and token-level credit assignment, nor proposed an adaptation-aware intrinsic weighting
 98 scheme.

99 2.2 Positioning of This Work

100 The literature makes three points clear. First, critics are not strictly required for effective LLM post-
 101 training: critic-free estimators such as RLOO, ReMax, GRPO, and GSPO are already competitive in
 102 both RLHF and RLVR [1, 19, 29, 43]. Second, many researchers have concluded that trajectory-level
 103 rewards are too coarse and have introduced denser supervision through process reward models,
 104 redistribution rules, tree structures, optimal baselines, temporal traces, or entropy-based modulation
 105 [16, 7, 32, 3, 24, 18, 13, 9, 41, 20]. Third, the overwhelming majority of practical LLM-RL pipelines
 106 are *trained with LoRA*, and a focused sub-literature has studied PEFT-specific effects in this regime [11,
 107 33, 39, 2, 15, 21, 17].

108 Our contribution is to bridge the second and third of these threads. Existing intrinsic credit-assignment
 109 signals (surprisal, entropy, divergence) are not novel as standalone objects, and we do not claim
 110 otherwise; they are our baselines. Our novel claim is that these signals *interact pathologically with*
 111 *the adaptation strategy that the field has converged on*: under LoRA they collapse toward uniform,
 112 so any paper that reports a null result for “fancy intrinsic weighting versus uniform” under LoRA is
 113 observing a LoRA artefact, not evidence against token-level credit assignment. We formalize this,
 114 supply a unified diagnostic (Gini/EffN) that makes the collapse visible at training time, and propose
 115 ARCA as an adaptation-aware alternative whose construction directly avoids the failure mode we
 116 identify.

117 3 Methods

118 We consider on-policy reinforcement learning for autoregressive language models with trajectory-
 119 level rewards. Let π_θ denote a language model parameterized by θ , generating a completion $y =$
 120 $(y_1, \dots, y_T)$ conditioned on a prompt x . After sampling a full trajectory, the model receives a scalar
 121 reward $R(x, y) \in \mathbb{R}$ computed by an external verifier, such as exact-answer correctness or unit-test
 122 pass rate. Our objective is

$$J(\theta) = \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_\theta(\cdot|x)}[R(x, y)], \quad (1)$$

123 where $\mathcal{D}$ denotes the prompt distribution. We focus on the regime most relevant to recent reasoning-
 124 model training: sparse outcome rewards, on-policy sampling, and no learned value function.

125 3.1 Policy Gradient with Trajectory-Level Reward

126 For a fixed prompt x , the standard REINFORCE estimator is

$$g_{\text{rf}}(x, y) = R(x, y) \sum_{t=1}^T \nabla_\theta \log \pi_\theta(y_t \mid y_{<t}, x). \quad (2)$$

127 In practice, a baseline $b(x)$ is subtracted to reduce variance without changing the expected gradient
 128 as long as $b(x)$ does not depend on the sampled action at token t :

$$g_{\text{base}}(x, y) = (R(x, y) - b(x)) \sum_{t=1}^T \nabla_\theta \log \pi_\theta(y_t \mid y_{<t}, x). \quad (3)$$

We use prompt-level multi-sample baselines. Given K sampled completions $\{y^{(i)}\}_{i=1}^K$ for the same prompt x with rewards $\{R^{(i)}\}_{i=1}^K$:

$$b_{\text{RLOO}}^{(i)}(x) = \frac{1}{K-1} \sum_{j \neq i} R^{(j)} \quad (4)$$

is the leave-one-out baseline used in RLOO-style estimators, while GRPO-style normalization can be written as

$$A_{\text{GRPO}}^{(i)}(x) = \frac{R^{(i)} - \mu_R(x)}{\sigma_R(x) + \varepsilon}, \quad (5)$$

where $\mu_R(x)$ and $\sigma_R(x)$ are the within-prompt mean and standard deviation of the K rewards. In both cases, the resulting scalar advantage is then broadcast uniformly over all tokens in the sampled trajectory. This scalar-broadcast assumption is precisely what we relax.

3.2 Token-Level Credit Redistribution

We introduce token-level weights $w_t(x, y)$ that redistribute a trajectory-level advantage across tokens:

$$g_w(x, y) = A(x, y) \sum_{t=1}^T w_t(x, y) \nabla_{\theta} \log \pi_{\theta}(y_t \mid y_{<t}, x), \quad (6)$$

where $A(x, y)$ denotes either $R(x, y) - b_{\text{RLOO}}(x)$ or $A_{\text{GRPO}}(x, y)$ depending on the baseline choice.

The weights satisfy

$$\sum_{t=1}^T w_t(x, y) = 1, \quad w_t(x, y) \geq 0. \quad (7)$$

This normalization keeps the total update magnitude comparable across weighting schemes and isolates the effect of credit redistribution from simple rescaling.

Uniform token credit corresponds to $w_t = 1/T$ for all t . Our goal is to replace this uniform allocation with *intrinsic* weights derived from quantities already produced by the policy during generation. We do not train token-level reward models, estimate prefix values, or build explicit search trees.

3.3 Intrinsic Token-Weighting Mechanisms

We define each method through an unnormalized salience score $\alpha_t(x, y) \geq 0$ and then normalize within each trajectory:

$$w_t(x, y) = \frac{\alpha_t(x, y) + \varepsilon}{\sum_{k=1}^T (\alpha_k(x, y) + \varepsilon)}, \quad (8)$$

with a small $\varepsilon > 0$ to avoid degenerate all-zero cases. In implementation, the weights are treated as *detached* scalars when multiplying the policy-gradient term, so the update does not introduce second-order derivatives through the weighting function itself. This keeps the optimization rule close in spirit and cost to standard critic-free policy gradients.

Uniform Weighting (Baseline).

$$w_t^{\text{uniform}} = \frac{1}{T}. \quad (9)$$

Combined with an RLOO or GRPO baseline, this recovers the standard critic-free estimator that assigns identical credit to every token in the sampled completion.

Surprisal Weighting. Our first intrinsic score is token surprisal,

$$\alpha_t^{\text{surp}}(x, y) = -\log \pi_{\theta}(y_t \mid y_{<t}, x). \quad (10)$$

This emphasizes low-probability decisions under the current policy. Intuitively, these are tokens where the model commits to a less routine continuation and where uniform credit assignment may be most diluted by predictable filler tokens.

158 **Entropy-Reduction Weighting.** Our second intrinsic score measures local entropy reduction. Let

$$H_t^{\text{pre}}(x, y) = - \sum_v \pi_\theta(v \mid y_{<t}, x) \log \pi_\theta(v \mid y_{<t}, x) \quad (11)$$

159 denote the predictive entropy before sampling token y_t , and let

$$\Delta H_t^{\text{ent}}(x, y) = \max(0, H_t^{\text{pre}}(x, y) - H_{t+1}^{\text{pre}}(x, y)) \quad (12)$$

160 for $t < T$. For the final token we set $\Delta H_T^{\text{ent}} = 0$. We then use

$$\alpha_t^{\text{ent}}(x, y) = \Delta H_t^{\text{ent}}(x, y). \quad (13)$$

161 This score highlights tokens after which the model’s next-step distribution becomes substantially
162 more concentrated. Such events often correspond to commitment points in reasoning, where one
163 branch of continuation becomes much more likely than its alternatives.

164 **Policy-Divergence Weighting.** A third intrinsic score measures how much the current policy has
165 drifted from a reference π_{ref} (typically the base model prior to RL) at each token position:

$$\alpha_t^{\text{div}}(x, y) = |\log \pi_\theta(y_t \mid y_{<t}, x) - \log \pi_{\text{ref}}(y_t \mid y_{<t}, x)|. \quad (14)$$

166 In the RLHF/RLVR setting π_{ref} is already available because it is used to compute the KL regularizer,
167 so this score is essentially free.

168 **Length-Robust Normalization.** Because reasoning trajectories can vary substantially in length,
169 all weights are normalized within each sampled completion rather than across the minibatch. This
170 ensures that longer trajectories do not automatically receive larger aggregate updates simply because
171 they contain more token positions with nonzero salience. The comparison between weighting schemes
172 therefore reflects *where* credit is assigned within a trajectory, not how much total credit a longer
173 sequence receives.

174 **Batch-Level Baselines.** All weighting schemes can be paired with either RLOO or GRPO-style
175 prompt-level baselines. Unless otherwise specified, we use the same rollout groups, optimizer,
176 sampling hyperparameters, and baseline family across methods. This isolates token redistribution as
177 the only algorithmic difference.

178 3.4 Signal Collapse Under Low-Rank Adaptation

179 So far our exposition has treated the policy π_θ as an unconstrained language model being updated
180 by gradient descent. In practice, however, almost all LLM-RL pipelines apply LoRA [11], which
181 restricts π_θ to a rank- r perturbation of a frozen reference policy π_{ref} . We now show that the intrinsic
182 weighting schemes defined above are qualitatively degraded by this constraint.

183 **Notation.** Write the logits produced by the base model at position t as $z_t^{\text{ref}} = W_{\text{lm}} h_t^{\text{base}}$ and the
184 logits produced by the adapted model as $z_t^\theta = W_{\text{lm}} h_t^{\text{adapted}}$, where $h_t^{\text{adapted}} = h_t^{\text{base}} + \Delta h_t$ and
185 Δh_t is the sum of LoRA adapter contributions propagated through the transformer to position t . Each
186 adapter contributes $B_\ell A_\ell x_{\ell,t}$ at layer ℓ , where $B_\ell \in \mathbb{R}^{d \times r}$ and $A_\ell \in \mathbb{R}^{r \times d}$, so the raw per-layer
187 perturbation at each position lies in the fixed r -dimensional column space of B_ℓ .

188 **Collapse of surprisal and entropy.** Because $\|\Delta h_t\|$ is bounded by a product of adapter norms and
189 input activation norms that are themselves roughly stationary across positions in a well-trained base
190 model, the first-order perturbation to per-token surprisal,

$$-\log \pi_\theta(y_t \mid y_{<t}, x) = -\log \pi_{\text{ref}}(y_t \mid y_{<t}, x) + O(\|\Delta h_t\|),$$

191 is dominated by the base model’s surprisal pattern, which is task-agnostic and primarily reflects filler
192 versus content tokens. The same holds for predictive entropy. Consequently, when we normalize
193 these scores within a trajectory (equation (8)), the resulting weights are nearly identical to the base
194 model’s surprisal or entropy weights, regardless of what the adapter has learned.

195 **Collapse of policy divergence.** Divergence weighting, $\alpha_t^{\text{div}} = |\log \pi_\theta(y_t) - \log \pi_{\text{ref}}(y_t)|$, is
 196 of direct interest because it quantifies exactly the policy change that the RL update is trying to
 197 drive. Under LoRA, however, this quantity is small and approximately uniform across positions
 198 for a different reason: the KL budget $\text{KL}(\pi_\theta \parallel \pi_{\text{ref}})$ is bounded by the rank- r adapter’s capacity, so
 199 the per-token log-probability ratios are compressed toward zero [2]. Under the within-trajectory
 200 normalization in equation (8), this compression maps any approximately constant score function to
 201 the uniform distribution, giving $w_t^{\text{div}} \approx 1/T$.

202 **Consequence.** We summarize the practical implication as follows. Let the Gini coefficient $G(w)$
 203 and the effective-token ratio $\text{EffN}(w)/T = 1/(T \sum_t w_t^2)$ be standard concentration measures, with
 204 $G(w) = 0$ and $\text{EffN}/T = 1$ corresponding to perfectly uniform weights. Then for surprisal,
 205 entropy-reduction, and divergence weighting under LoRA, both concentration measures approach
 206 their uniform limits as the adapter rank r decreases relative to the hidden dimension d . In this regime,
 207 “intrinsic token weighting” is indistinguishable from uniform broadcast, and intrinsic-weighting
 208 methods cannot provide the credit-redistribution benefit they were designed to deliver. This is *not* a
 209 problem with the underlying signals; it is a structural consequence of applying them inside a low-rank
 210 adaptation.

211 3.5 Adapter-Residual Credit Assignment (ARCA)

212 The collapse result motivates a different design choice. Rather than measuring per-token properties
 213 of the *output distribution* (whose shape is dominated by the frozen base model under LoRA), we
 214 measure per-token properties of the *adapter’s own contribution* to the forward pass.

215 **Definition.** Given a model with a LoRA adapter, let h_t^{adapted} denote the last-layer hidden state at
 216 position t with the adapter enabled, and h_t^{base} the same hidden state with the adapter disabled. Define
 217 the Adapter-Residual Credit Assignment salience as

$$\alpha_t^{\text{ARCA}}(x, y) = \|h_t^{\text{adapted}} - h_t^{\text{base}}\|_2, \quad (15)$$

218 and normalize within the trajectory as in equation (8) to obtain per-token weights w_t^{ARCA} . The
 219 adapter residual is computed once per minibatch via an extra no-grad forward pass with the adapter
 220 disabled, using `model.disable_adapter()` in frameworks such as PEFT. This is exactly the same
 221 call already required to compute a KL regularizer against the reference policy or to support divergence
 222 weighting, so ARCA adds no new infrastructure.

223 **Why ARCA does not collapse.** The key property that distinguishes ARCA from the intrinsic
 224 schemes above is that it measures a quantity whose *per-token* value is actively shaped by the attention
 225 and MLP pathways through which the adapter input activations are routed. Even for a low-rank
 226 adapter with small $\|B_\ell A_\ell\|$, the input activations $x_{\ell,t}$ are non-uniform across positions (attention
 227 patterns, layer norms, and content-versus-filler distinctions are already non-uniform in the base
 228 model), so $\|\Delta h_t\|$ retains non-trivial position-to-position variation. Formally, as the adapter rank
 229 goes to zero the signal α_t^{ARCA} vanishes uniformly, but its *normalized* counterpart w_t^{ARCA} remains
 230 non-degenerate as long as the adapter is nontrivially active at *any* position.

231 **Interpretation as implicit gradient-informed credit.** ARCA can be read as a cheap proxy for a
 232 gradient-based notion of token importance. Positions at which the adapter contribution is large are
 233 positions at which the adapter’s parameter gradient has high contribution to the loss, and therefore
 234 positions where a policy-gradient update can have the most effect. This is a substantially weaker
 235 statement than the one made by a learned critic, but it has the advantage of being available for free
 236 from quantities the adapted forward pass already computes, and of being exactly targeted at the
 237 PEFT-specific failure mode introduced above.

238 4 Experiments

239 We evaluate whether adapter-residual credit assignment gives a useful token-level signal in the
 240 low-rank fine-tuning regime. Because the submission-time experiments are intentionally compact, the
 241 empirical section focuses on a single controlled sweep: one model, one task, one advantage estimator,

Table 1: **Submission-time sweep.** All runs use Qwen/Qwen3-1.7B-Base, MATH, GRPO, one seed, and 100 training steps. The non-ARCA baselines are matched at LoRA rank 64; ARCA is evaluated at three ranks to separate credit assignment from adapter capacity.

Method	LoRA rank	Trainable params	Role in comparison
Uniform	64	69.7M	Same-rank baseline
Surprisal	64	69.7M	Intrinsic token baseline
Entropy reduction	64	69.7M	Intrinsic token baseline
Policy divergence	64	69.7M	Intrinsic token baseline
ARCA	4	4.36M	Low-rank ARCA control
ARCA	16	17.4M	Mid-rank ARCA control
ARCA	64	69.7M	Same-rank ARCA comparison

and seven weighting configurations. This design trades breadth for a cleaner test of the central claim: ARCA changes credit assignment without adding trainable parameters beyond the LoRA adapter used by the matched baselines.

4.1 Setup

Model and task. We fine-tune Qwen/Qwen3-1.7B-Base on the MATH training split and evaluate on held-out MATH examples. Prompts ask the model to solve the problem step by step and place the final answer in `\boxed{}`. Rewards are binary exact-match scores computed from the final boxed answer after normalization.

Training. All runs use GRPO prompt-level advantages, LoRA adapters on the attention and MLP projections, and the same rollout budget, optimizer, learning-rate schedule, prompt format, and decoding parameters. Each training step samples $K = 4$ completions per prompt, uses 100 update steps, and evaluates greedy accuracy and pass@4 on 200 held-out MATH examples. We run a single seed, 1337. Checkpoints are saved every 20 steps. We use AdamW with learning rate 5×10^{-6} , zero weight decay, a 10-step warmup followed by cosine decay, batch size 2, gradient accumulation over 4 microbatches, maximum prompt length 512, maximum generation length 512, temperature 1.0, and top- p 0.95 sampling during training.

Methods. The sweep contains seven runs. Four baselines use LoRA rank $r = 64$: uniform token weighting, surprisal weighting, entropy-reduction weighting, and policy-divergence weighting. The proposed method, ARCA, is run at $r \in \{4, 16, 64\}$. The $r = 64$ comparison tests ARCA against baselines with the same trainable parameter count. The $r = 4$ and $r = 16$ comparisons test whether ARCA remains competitive with fewer adapter parameters than the $r = 64$ baselines.

4.2 Main Results on MATH

Table 2 reports held-out MATH performance after RL fine-tuning. The primary comparison is ARCA at rank 64 versus the rank-64 baselines, which holds the trainable parameter count fixed. The rank-4 and rank-16 ARCA runs provide a more stringent control: if lower-rank ARCA is competitive with rank-64 baselines, the gain cannot be attributed simply to using a larger adapter.

Table 2 should be read as a compact validation rather than a broad benchmark. Uniform weighting obtains the highest greedy accuracy, while entropy reduction obtains the highest pass@4. ARCA at rank 64 is competitive with these baselines under the same trainable parameter count, trailing uniform greedy accuracy by three points and exceeding uniform pass@4 by 0.5 points. The lower-rank ARCA variants do not outperform the rank-64 baselines in this run, which is consistent with a capacity-performance tradeoff rather than evidence that ARCA gains come from extra parameters.

4.3 Credit-Assignment Diagnostics

Performance alone does not show whether a method changes token-level credit assignment. We therefore log the concentration of token weights during training. We report the Gini coefficient

Table 2: **Held-out MATH performance.** Greedy accuracy and pass@4 after GRPO fine-tuning. ARCA at rank 64 is parameter-matched to the rank-64 baselines; lower-rank ARCA rows test the parameter-count alternative. Train reward is averaged over the 100 update steps.

Method	LoRA rank	Greedy acc.	Pass@4	Avg. train reward
Uniform	64	0.620	0.675	0.362
Surprisal	64	0.520	0.650	0.326
Entropy reduction	64	0.600	0.700	0.367
Policy divergence	64	0.565	0.670	0.351
ARCA	4	0.525	0.640	0.327
ARCA	16	0.510	0.655	0.336
ARCA	64	0.590	0.680	0.353

Table 3: **Token-weight diagnostics.** Concentration statistics averaged over training. These diagnostics test whether a method produces a non-uniform but non-collapsed token-credit signal, the empirical observable predicted by the collapse analysis.

Method	LoRA rank	$G(w)$	EffN/T
Uniform	64	0.000	1.000
Surprisal	64	0.941	0.052
Entropy reduction	64	0.920	0.073
Policy divergence	64	0.939	0.055
ARCA	4	0.353	0.467
ARCA	16	0.388	0.460
ARCA	64	0.490	0.334

277 $G(w)$ and effective-token ratio $\text{EffN}(w)/T$. Uniform weighting has $G = 0$ and $\text{EffN}/T = 1$; highly
278 concentrated weights have G near one and a small effective-token ratio.

279 Table 3 is the central diagnostic result. Uniform weighting is exactly flat. Surprisal, entropy reduction,
280 and policy divergence produce extremely concentrated distributions, with effective-token ratios
281 between 0.052 and 0.073 on average. ARCA occupies the predicted middle regime: it is far from
282 uniform, but it does not collapse onto the tiny effective support used by the output-distribution
283 baselines. This is the empirical signature of the theoretical distinction. Output-distribution scores ask
284 where the model is uncertain or shifted; ARCA asks where the adapter actually acts.

285 4.4 Results and Discussion

286 The results support a theory-first interpretation. The downstream accuracy numbers do not show a
287 universal ARCA win: uniform is best under greedy decoding, and entropy reduction is best under
288 pass@4. However, ARCA at rank 64 is competitive under matched parameters, and its token-credit
289 distribution is qualitatively different from every baseline. In this paper, that diagnostic is not secondary.
290 It is the observable predicted by the LoRA-collapse analysis: a method whose salience is measured
291 through output distributions either broadcasts credit uniformly or concentrates it on a very small
292 set of positions, while adapter-residual salience remains position-discriminative without becoming
293 maximally sparse.

294 This distinction matters because token-level credit assignment is often evaluated only by final task
295 accuracy. Accuracy alone cannot tell whether a method actually changed token-level credit or merely
296 produced a different optimization trajectory. The concentration diagnostics separate these cases.
297 ARCA demonstrates that one can obtain an adaptation-aware credit signal from the LoRA adapter
298 itself, with no value head, process reward model, or tree construction, while keeping the trainable
299 parameter count identical to the rank-64 baselines.

300 The rank controls provide a useful boundary on the claim. ARCA at ranks 4 and 16 uses substantially
301 fewer trainable parameters than the rank-64 baselines, and in this sweep those lower-rank variants
302 do not match rank-64 performance. We therefore do not claim that ARCA removes the need for
303 sufficient adapter capacity. The supported claim is sharper: at a fixed adapter rank, ARCA changes the

304 geometry of token credit in the way predicted by the theory, and does so while remaining competitive
305 on held-out MATH.

306 4.5 Implementation and Metrics

307 The main experiment entrypoint is `experiment/hf/run_experiment.py`. For each minibatch, the
308 script samples completions from the current policy, computes binary rewards after full generation,
309 constructs GRPO advantages, and applies the weighted token-level objective from Section 3. Surprisal
310 uses current-policy token log probabilities, entropy-reduction uses next-token entropy drops, policy-
311 divergence compares adapted and adapter-disabled log probabilities, and ARCA weights tokens by
312 adapter-residual norms in the final hidden layer.

313 The code records per-step reward mean, reward variance, loss, completion length, token-weight
314 Gini, effective-token ratio, and, for ARCA, the mean and maximum adapter-residual norm. Final
315 evaluation records greedy accuracy and pass@4 on held-out MATH examples. These metrics allow
316 us to distinguish three claims: whether the method trained, whether it changed token-credit structure,
317 and whether that structure improved held-out performance.

318 4.6 Compute

319 Each run fits on a single H100-class GPU. The seven submission-time runs are executed independently,
320 one configuration per GPU, with wall-clock training time on the order of a few hours per run plus final
321 evaluation. The reported sweep therefore requires seven single-GPU runs; preliminary development
322 runs were used to tune runtime and checkpointing but are not included in the reported comparison.

323 4.7 Broader Impacts

324 This work is methodological and studies how to assign token-level credit during reinforcement
325 learning for language models. The potential positive impact is improved sample efficiency and
326 interpretability of RL fine-tuning, especially in settings where practitioners already use parameter-
327 efficient adaptation. The same techniques could also improve the fine-tuning of models for harmful
328 or misleading generation if applied without appropriate task, data, and deployment safeguards. We
329 do not release a new general-purpose model in this paper.

330 4.8 Limitations

331 This sweep is deliberately small: one model, one dataset, one seed, one advantage estimator, and
332 seven configurations. It is sufficient to test the parameter-matched ARCA comparison and the low-
333 rank control, but it does not establish robustness across seeds, tasks, or algorithms. We view the
334 results as a targeted validation of the mechanism rather than a broad benchmark claim.

335 5 Conclusion

336 Token-level credit assignment and parameter-efficient fine-tuning are usually studied as separate
337 design choices. This paper argues that they are coupled. Under LoRA, output-distribution signals
338 such as surprisal, entropy reduction, and policy divergence can lose the token-level variation that
339 credit assignment methods rely on. ARCA addresses this failure mode by measuring salience
340 where LoRA actually acts: in the adapter-induced hidden-state residual. The resulting signal is
341 lightweight, requires no learned critic or process reward model, and produces non-collapsed token-
342 credit distributions in the regime where output-distribution baselines become either flat or highly
343 concentrated. Our MATH/Qwen3-1.7B sweep is intentionally compact, but it supports the central
344 mechanism: adaptation-aware credit assignment changes the geometry of token updates under
345 matched rollout and parameter budgets.

346 References

- 347 [1] Arash Ahmadian, Chris Cremer, Matthias Gallé, Marzieh Fadaee, Julia Kreutzer, Olivier
348 Pietquin, Ahmet Üstün, and Sara Hooker. Back to basics: Revisiting REINFORCE-style
349 optimization for learning from human feedback in LLMs. In *Proceedings of the 62nd Annual*

- 350 *Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association
351 for Computational Linguistics, 2024.
- 352 [2] Anonymous. LoRA as an implicit KL regularizer in GRPO fine-tuning: From theory to practice,
353 2026. Under review at Transactions on Machine Learning Research.
- 354 [3] Meng Cao, Shuyuan Zhang, Xiao-Wen Chang, and Doina Precup. SCAR: Shapley credit
355 assignment for more efficient RLHF, 2025.
- 356 [4] Yekun Chai, Haoran Sun, Huang Fang, Shuohuan Wang, Yu Sun, and Hua Wu. MA-RLHF: Re-
357 inforcement learning from human feedback with macro actions. In *The Thirteenth International
358 Conference on Learning Representations*, 2025.
- 359 [5] Hongzhan Chen, Tao Yang, Shiping Gao, Ruijun Chen, Xiaojun Quan, Hongtao Tian, and Ting
360 Yao. Discriminative policy optimization for token-level reward models. In Aarti Singh, Maryam
361 Fazel, Daniel Hsu, Simon Lacoste-Julien, Felix Berkenkamp, Tegan Maharaj, Kiri Wagstaff,
362 and Jerry Zhu, editors, *Proceedings of the 42nd International Conference on Machine Learning*,
363 volume 267 of *Proceedings of Machine Learning Research*, pages 9546–9565. PMLR, 2025.
- 364 [6] Paul F. Christiano, Jan Leike, Tom B. Brown, Miljan Martic, Shane Legg, and Dario Amodei.
365 Deep reinforcement learning from human preferences. In *Advances in Neural Information
366 Processing Systems 30*, 2017.
- 367 [7] Ganqu Cui, Lifan Yuan, Zefan Wang, Hanbin Wang, Yuchen Zhang, Jiacheng Chen, Wendi
368 Li, Bingxiang He, Yuchen Fan, Tianyu Yu, Qixin Xu, Weize Chen, Jiarui Yuan, Huayu Chen,
369 Kaiyan Zhang, Xingtai Lv, Shuo Wang, Yuan Yao, Xu Han, Hao Peng, Yu Cheng, Zhiyuan Liu,
370 Maosong Sun, Bowen Zhou, and Ning Ding. Process reinforcement through implicit rewards,
371 2025.
- 372 [8] DeepSeek-AI. Deepseek-r1: Incentivizing reasoning capability in LLMs via reinforcement
373 learning, 2025.
- 374 [9] Yuhang He, Haodong Wu, Siyi Liu, Hongyu Ge, Hange Zhou, Keyi Wu, Zhuo Zheng, Qihong
375 Lin, Zixin Zhong, and Yongqi Zhang. Rethinking token-level credit assignment in RLVR: A
376 polarity-entropy analysis, 2026.
- 377 [10] Falko Helm, Nico Daheim, and Iryna Gurevych. Token weighting for long-range language
378 modeling. In Luis Chiruzzo, Alan Ritter, and Lu Wang, editors, *Findings of the Association for
379 Computational Linguistics: NAACL 2025*, pages 1440–1459, Albuquerque, New Mexico, April
380 2025. Association for Computational Linguistics.
- 381 [11] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yanzhi Li, Shean Wang,
382 Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In
383 *International Conference on Learning Representations*, 2022.
- 384 [12] Jian Hu, Jason Klein Liu, Haotian Xu, and Wei Shen. REINFORCE++: Stabilizing critic-free
385 policy optimization with global advantage normalization, 2025.
- 386 [13] Miaobo Hu, BoKun Wang, Shuhao Hu, Ruohan Wang, Xin Wang, Xiaobo Guo, Daren Zha,
387 and Jun Xiao. PSPO: Trainable potential-based reward shaping with internal model signals
388 for post-training policy optimization of large language models, 2026. ICLR 2026 submission,
389 rejected.
- 390 [14] Amirhossein Kazemnejad, Milad Aghajohari, Eva Portelance, Alessandro Sordoni, Siva Reddy,
391 Aaron Courville, and Nicolas Le Roux. VinePPO: Refining credit assignment in RL training of
392 LLMs, 2024.
- 393 [15] Alan Lee and Harry Tong. Token-efficient RL for LLM reasoning, 2025.
- 394 [16] Jiahui Li, Lin Li, Tai-Wei Chang, Kun Kuang, Long Chen, Jun Zhou, and Cheng Yang. Red:
395 Unleashing token-level rewards from holistic feedback via reward redistribution, 2024.
- 396 [17] Shiwei Li, Xiandi Luo, Haozhao Wang, Xing Tang, Ziqiang Cui, Dugang Liu, Yuhua Li,
397 Xiuqiang He, and Ruixuan Li. Beyond higher rank: Token-wise input-output projections for
398 efficient low-rank adaptation, 2025.

- [18] Yingru Li, Jiawei Xu, Ziniu Li, Jiakai Liu, Wei Liu, Yuxuan Tong, Longtao Zheng, Zhenghai Xue, Yaxiang Zhang, Tianle Cai, Ge Zhang, Qian Liu, and Baoxiang Wang. The optimal token baseline: Variance reduction for long-horizon LLM-RL, 2026.
- [19] Ziniu Li, Tian Xu, Yushun Zhang, Zhihang Lin, Yang Yu, Ruoyu Sun, and Zhi-Quan Luo. ReMax: A simple, effective, and efficient reinforcement learning method for aligning large language models. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*. PMLR, 2024.
- [20] Haoming Meng, Kexin Huang, Shaohang Wei, Chiyu Ma, Shuo Yang, Xue Wang, Guoyin Wang, Bolin Ding, and Jingren Zhou. Sparse but critical: A token-level analysis of distributional shifts in RLVR fine-tuning of LLMs, 2026.
- [21] Julian Minder, Clément Dumas, Stewart Slocum, Helena Casademunt, Cameron Holmes, Robert West, and Neel Nanda. Narrow finetuning leaves clearly readable traces in activation differences, 2025.
- [22] Youssef Mroueh, Nicolas Dupuis, Brian Belgodere, Apoorva Nitsure, Mattia Rigotti, Kristjan Greenewald, Jiri Navratil, Jerret Ross, and Jesus Rios. Revisiting group relative policy optimization: Insights into on-policy and off-policy training, 2025.
- [23] Long Ouyang, Jeff Wu, Xu Jiang, Diogo Almeida, Carroll L. Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul Christiano, Jan Leike, and Ryan Lowe. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems 35*, 2022.
- [24] Prasanna Parthasarathi, Mathieu Reymond, Boxing Chen, Yufei Cui, and Sarath Chandar. Grpo- λ : Credit assignment improves llm reasoning, 2025.
- [25] John Schulman, Sergey Levine, Pieter Abbeel, Michael Jordan, and Philipp Moritz. Trust region policy optimization. In Francis Bach and David Blei, editors, *Proceedings of the 32nd International Conference on Machine Learning*, volume 37 of *Proceedings of Machine Learning Research*, pages 1889–1897, Lille, France, 07–09 Jul 2015. PMLR.
- [26] John Schulman, Philipp Moritz, Sergey Levine, Michael Jordan, and Pieter Abbeel. High-dimensional continuous control using generalized advantage estimation. In *International Conference on Learning Representations*, 2016.
- [27] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms, 2017.
- [28] Zikang Shan, Han Zhong, Liwei Wang, and Li Zhao. Bringing value models back: Generative critics for value modeling in LLM reinforcement learning, 2026.
- [29] Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. Deepseekmath: Pushing the limits of mathematical reasoning in open language models, 2024.
- [30] Lloyd S. Shapley. A value for n-person games. In Harold W. Kuhn and Albert W. Tucker, editors, *Contributions to the Theory of Games II*, pages 307–317. Princeton University Press, Princeton, 1953.
- [31] Nisan Stiennon, Long Ouyang, Jeffrey Wu, Daniel M. Ziegler, Ryan Lowe, Chelsea Voss, Alec Radford, Dario Amodei, and Paul F. Christiano. Learning to summarize from human feedback. In *Advances in Neural Information Processing Systems 33*, 2020.
- [32] Hieu Tran, Zonghai Yao, and Hong Yu. Exploiting tree structure for credit assignment in RL training of LLMs, 2025.
- [33] Shangshang Wang, Julian Asilis, Ömer Faruk Akgül, Enes Burak Bilgin, Ollie Liu, and Willie Neiswanger. Tina: Tiny reasoning models via LoRA, 2025.

- [34] Shenzhi Wang, Le Yu, Chang Gao, Chujie Zheng, Shixuan Liu, Rui Lu, Kai Dang, Xiong-Hui Chen, Jianxin Yang, Zhenru Zhang, Yuqiong Liu, An Yang, Andrew Zhao, Yang Yue, Shiji Song, Bowen Yu, Gao Huang, and Junyang Lin. Beyond the 80/20 rule: High-entropy minority tokens drive effective reinforcement learning for LLM reasoning, 2025.
- [35] Xumeng Wen, Zihan Liu, Shun Zheng, Shengyu Ye, Zhirong Wu, Yang Wang, Zhijian Xu, Xiao Liang, Junjie Li, Ziming Miao, Jiang Bian, and Mao Yang. Reinforcement learning with verifiable rewards implicitly incentivizes correct reasoning in base LLMs, 2025.
- [36] Ronald J. Williams. Simple statistical gradient-following algorithms for connectionist reinforcement learning. *Machine Learning*, 8:229–256, may 1992.
- [37] Wei Xiong, Jiarui Yao, Yuhui Xu, Bo Pang, Lei Wang, Doyen Sahoo, Junnan Li, Nan Jiang, Tong Zhang, Caiming Xiong, and Hanze Dong. A minimalist approach to LLM reasoning: from rejection sampling to reinforce, 2025.
- [38] Shentao Yang, Shujian Zhang, Congying Xia, Yihao Feng, Caiming Xiong, and Mingyuan Zhou. Preference-grounded token-level guidance for language model fine-tuning. In Alice Oh, Tristan Naumann, Amir Globerson, Kate Saenko, Moritz Hardt, and Sergey Levine, editors, *Advances in Neural Information Processing Systems 36: Annual Conference on Neural Information Processing Systems 2023, NeurIPS 2023, New Orleans, LA, USA, December 10–16, 2023*, 2023.
- [39] Qingyu Yin, Yulun Wu, Zhennan Shen, Sunbowen Li, Zhilin Wang, Yanshu Li, Chak Tou Leong, Jiale Kang, and Jinjin Gu. Evaluating parameter efficient methods for RLVR, 2025.
- [40] Hee Suk Yoon, Eunseop Yoon, Mark A. Hasegawa-Johnson, Sungwoong Kim, and Chang D. Yoo. ConfPO: Exploiting policy model confidence for critical token selection in preference optimization. In Aarti Singh, Maryam Fazel, Daniel Hsu, Simon Lacoste-Julien, Felix Berkenkamp, Tegan Maharaj, Kiri Wagstaff, and Jerry Zhu, editors, *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 72641–72655. PMLR, 2025.
- [41] Song Yu, Li Li, Wenwen Zhao, and Zhisheng Yang. ERPO: Token-level entropy-regulated policy optimization for large reasoning models, 2026.
- [42] Chong Zhang, Yue Deng, Xiang Lin, Bin Wang, Dianwen Ng, Hai Ye, Xingxuan Li, Yao Xiao, Zhanfeng Mo, Qi Zhang, and Lidong Bing. 100 days after deepseek-r1: A survey on replication studies and more directions for reasoning language models, 2025.
- [43] Chujie Zheng, Shixuan Liu, Mingze Li, Xiong-Hui Chen, Bowen Yu, Chang Gao, Kai Dang, Yuqiong Liu, Rui Men, An Yang, Jingren Zhou, and Junyang Lin. Group sequence policy optimization, 2025.
- [44] Han Zhong, Zikang Shan, Guhao Feng, Wei Xiong, Xinle Cheng, Li Zhao, Di He, Jiang Bian, and Liwei Wang. DPO meets PPO: Reinforced token optimization for RLHF. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 78498–78521. PMLR, 2025.

A Extended Related Work

A.1 From RLHF to RLVR

Modern language-model post-training inherits its optimization machinery from policy-gradient reinforcement learning, from REINFORCE [36] to trust-region and clipped-surrogate methods such as TRPO and PPO [25, 27]. Advantage estimation, especially GAE, became the standard variance-reduction device in continuous-control RL by learning value functions over states or prefixes [26]. In language modeling, these ideas entered mainstream use through RLHF systems that optimize sequence-level rewards derived from human preferences or reward models [6, 31, 23].

The recent reasoning-model wave shifted part of the field from preference-based RLHF toward reinforcement learning with verifiable rewards (RLVR), where supervision is often a deterministic correctness signal on math or code tasks. DeepSeekMath introduced GRPO as a practical critic-free

alternative for these settings [29], and DeepSeek-R1 popularized large-scale RL-only or RL-dominant reasoning pipelines [8]. Subsequent surveys document how quickly this regime became the default template for open reasoning-model replication and extension [42]. This transition matters for our setting because RLVR makes sparse outcome rewards and long reasoning trajectories central, thereby exposing the credit-assignment problem more directly than earlier short-form RLHF tasks.

A.2 Critic-Based and Critic-Free Policy Optimization

The classical solution to delayed reward is to learn a critic or value function and convert returns into token- or prefix-level advantages. PPO-style RLHF pipelines follow this recipe, but the value-estimation problem is particularly awkward for text generation because prefixes are high-dimensional, non-Markov, and semantically heterogeneous. VinePPO demonstrated that standard value heads produce poor estimates of expected returns for reasoning tasks and proposed Monte Carlo vine-style rollout estimates as a more accurate alternative [14]. GenAC extends this idea by replacing scalar value prediction with a generative critic that uses chain-of-thought reasoning for value estimation [28].

A growing body of work questions whether value heads are necessary at all in LLM post-training. ReMax replaces learned critics with a greedy baseline and shows that much of PPO’s practical benefit can be retained with a simpler REINFORCE-style objective [19]. Back to Basics systematically compares PPO with RLOO-style estimators and finds that critic-free methods can match or exceed value-based baselines in RLHF [1]. REINFORCE++ further strengthens this line by replacing the prompt-level advantage normalization used by GRPO and RLOO with a global, batch-level normalization, stabilizing critic-free policy optimization without reintroducing a learned value function [12]. In reasoning-focused RLVR, GRPO normalizes rewards across a group of sampled responses rather than learning prefix values [29], and later work refines this family with theoretical analyses, off-policy variants, and sequence-level formulations such as GSPO [22, 43]. These methods substantially simplify the optimization stack, but they generally still broadcast a trajectory-level signal across all tokens once a scalar baseline is fixed.

A.3 Reasoning-Specific Analyses of RLVR

Another recent line asks why critic-free RLVR works so well for reasoning in the first place. A Minimalist Approach to LLM Reasoning shows that a simple rejection-sampling baseline (RAFT) is surprisingly competitive with GRPO and PPO on reasoning tasks, and that GRPO’s advantage over vanilla REINFORCE comes primarily from discarding prompts whose sampled responses are all incorrect rather than from reward normalization; the authors distill this insight into Reinforce-Rej, a minimal REINFORCE variant that filters both entirely-correct and entirely-incorrect samples, suggesting that much of the field’s complexity is optional rather than essential [37]. Complementarily, Wen et al. analyze RLVR theoretically and empirically, arguing that answer-level verifiable rewards can nonetheless incentivize correct intermediate reasoning early in a trajectory [35]. These results are important because they show that uniform outcome-level objectives already contain nontrivial reasoning signal.

At the same time, they do not eliminate the question of *which* tokens should absorb that signal. Recent empirical analysis of RLVR training dynamics suggests that improvements are disproportionately driven by a minority of high-entropy “forking” tokens, and that restricting updates to this subset can remain competitive or even outperform full-token updates in some settings [34]. Building on this observation, EAPO adapts conditional mutual information to the autoregressive RLVR setting and proves that the credit a token can carry is upper-bounded by its entropy, motivating an entropy-aware modulation of per-token learning signals [9]. ERPO similarly identifies “critical decision pivots” at high-entropy states and applies targeted exploration at those positions [41]. *Sparse but Critical* argues, via a distributional-shift analysis based on cross-sampling between policy and reference, that a small fraction of tokens accounts for most of the useful update magnitude in RLVR, and proposes a divergence-based selection rule for identifying them [20]. These entropy- and divergence-based methods are the closest concurrent work to the intrinsic weighting schemes that form our baselines; the present paper does *not* claim priority over them. Instead, our contribution is to show that their underlying signals are structurally degraded under LoRA, and to provide a credit-assignment mechanism (ARCA) that does not suffer the same degradation.

A.4 Fine-Grained Credit Assignment Beyond Uniform Token Updates

A large parallel literature attempts to make supervision denser than a single end-of-trajectory reward. One family learns explicit token- or process-level reward estimators. Preference-grounded Token-level Guidance derives token-level signals from preference data for fine-tuning [38], while Discriminative Policy Optimization learns token-level Q-style reward models from preferences without requiring fine-grained annotation [5]. PRIME pushes this direction further for reasoning tasks by constructing implicit process rewards and updating process reward models online from rollouts and outcome labels [7]. Reinforced Token Optimization (RTO) frames RLHF as a token-level MDP and uses DPO log-likelihood ratios as a per-token reward signal derived from the preference model, which is then optimized with PPO to produce fine-grained token-level credit [44]. These methods can produce rich token-level feedback, but they rely on an auxiliary reward-modeling component that our approach intentionally avoids.

A second family learns a small token-level critic on top of the policy’s own hidden states (rather than an external reward model) and uses its predictions to form token-level advantages. This corresponds closely to an earlier version of our own framework, and we found through an extensive literature review that the token-level actor–critic design space was already well-covered: VinePPO [14] and GenAC [28] cover non-parametric and generative critics respectively, and OTB [18] provides a principled variance-minimizing position-dependent baseline. We therefore do *not* position our critic-based configuration as a proposed method; we include it purely as a strong within-family baseline in our comparisons, and acknowledge this prior art explicitly.

A third family redistributes outcome rewards algorithmically rather than by training a dense reward model. RED derives token-level rewards by redistributing holistic reward-model scores over a trajectory [16]. SCAR uses Shapley-inspired attribution to estimate token or span contributions from sequence-level feedback [3, 30]. TEMPO uses groups of sampled solutions to build a prefix tree and compute nonparametric prefix values, yielding branch-sensitive corrections without a learned critic [32]. GRPO- λ introduces eligibility traces and lambda-returns into critic-free RLVR to propagate outcome information backward along the sequence [24]. OTB derives an optimal position-dependent baseline that minimizes per-token gradient variance; the practical estimator sets the baseline at position t to a cumulative-gradient-energy-weighted average of group returns-to-go, where the weights are a causal logit-gradient proxy computed directly from the forward-pass probabilities [18]. PSPO applies potential-based reward shaping theory to construct dense token rewards from outcome signals without altering the optimal policy [13]. Compared with these approaches, our focus is orthogonal: we do not propose a new redistribution rule, but instead diagnose a failure mode that affects essentially all intrinsic-signal-based redistribution methods when applied inside LoRA.

A.5 Alternative Action Granularities and Token Selection

Several papers attack the same underlying problem by changing the optimization unit rather than by redesigning the reward. MA-RLHF introduces macro actions, grouping tokens into higher-level units to shorten the temporal distance between decisions and rewards [4]. GSPO moves importance weighting and clipping from the token level to the sequence level for greater stability in large-scale RL training [43]. These methods reinforce the broader point that the granularity of optimization matters as much as the reward definition.

Outside on-policy RL proper, token-selective optimization has also been explored in adjacent paradigms. ConfPO emphasizes preference-critical tokens based on policy confidence in preference optimization [40], and token weighting for long-range language modeling shows that non-uniform token emphasis can improve optimization even in supervised settings [10]. Together with high-entropy token analyses in RLVR [34], these results strongly suggest that uniform token treatment is an arbitrary design choice rather than a necessity.

B Additional Method Context

B.1 Relationship to Existing Methods

Our formulation recovers standard critic-free RL as a special case: uniform weighting plus a prompt-level batch baseline yields the usual RLOO/GRPO-style estimator. The intrinsic weighting schemes

(surprisal, entropy reduction, policy divergence) are the natural baselines within our framework and map cleanly to published methods (e.g., divergence weighting corresponds to the divergence-advantage analyses of *Sparse but Critical* [20], entropy-based weighting to [9, 41]). ARCA is the only scheme we study that uses an explicitly *adaptation-aware* signal. Unlike critic-based PPO, ARCA does not learn a value head. Unlike RED, PRIME, or token-level reward-model methods, ARCA does not construct dense rewards from an auxiliary model. Unlike TEMPO or related tree-based methods, ARCA does not build prefix graphs or estimate branch values. Unlike hard token-selection methods based on top-entropy masking, ARCA retains a dense token update but modulates it continuously using an intrinsic salience score derived from the adapter itself.

The resulting estimator is intentionally minimal. It changes only the within-trajectory allocation of a scalar on-policy advantage, using quantities already available from the forward pass with the adapter disabled. This makes it easy to layer on top of existing critic-free RLVR pipelines while preserving their computational simplicity.

C Theoretical Interpretation

We provide an interpretation of intrinsic token weighting as a variance-control and credit-redistribution mechanism for policy gradient estimation in language models. Our goal is not to derive new convergence guarantees, but to clarify how token weighting relates to advantage estimation and why it can be effective without learning value functions.

C.1 Token Weighting as Credit Redistribution

Consider the standard REINFORCE estimator with a batch-level baseline:

$$g = (R - b) \sum_{t=1}^T \nabla_{\theta} \log \pi_{\theta}(y_t \mid y_{<t}, x). \quad (16)$$

This estimator assigns equal credit to all tokens in a trajectory. Introducing token weights yields:

$$g_w = (R - b) \sum_{t=1}^T w_t(y) \nabla_{\theta} \log \pi_{\theta}(y_t \mid y_{<t}, x), \quad (17)$$

where $\sum_t w_t = 1$.

Importantly, this transformation does not change the reward signal itself; it redistributes how the trajectory-level signal is attributed to individual decisions. When w_t depends only on quantities available at sampling time (e.g., log-probabilities or entropy), the estimator remains on-policy and does not require learning additional functions.

C.2 Relationship to Advantage Estimation

Advantage-based methods can be interpreted as implicitly defining token-level weights via a learned value function:

$$A_t = R - V(y_{<t}, x). \quad (18)$$

In this view, the contribution of each token is scaled according to how much the observed outcome deviates from an estimated expectation conditioned on the prefix.

However, this interpretation relies on the existence of a meaningful value function over prefixes. In language generation, prefixes are non-Markov and do not form a stable state space, making $V(y_{<t}, x)$ difficult to estimate and prone to bias. Token weighting offers an alternative: rather than estimating expected returns from prefixes, it emphasizes tokens based on intrinsic measures of decision salience, such as uncertainty or information gain.

C.3 Connection to Control Variates

From a statistical perspective, subtracting a baseline b is a form of control variate that reduces variance without affecting unbiasedness. Token weighting can be viewed as a complementary mechanism that

637 reshapes the contribution of individual score-function terms:

$$\nabla_{\theta} \log \pi_{\theta}(y) = \sum_{t=1}^T \nabla_{\theta} \log \pi_{\theta}(y_t \mid y_{<t}, x). \quad (19)$$

638 Uniform weighting treats all score-function terms equally, regardless of their variance or relevance.
 639 Intrinsic weighting schemes effectively reweight these terms, down-weighting low-variance or low-
 640 impact contributions (e.g., high-probability continuation tokens) and emphasizing high-variance or
 641 high-impact decisions (e.g., low-probability or uncertainty-reducing tokens). While this reweighting
 642 introduces bias relative to the uniform estimator, it can substantially reduce variance, leading to
 643 improved optimization dynamics in practice.

644 C.4 Interpretation of Specific Weighting Schemes

645 **Surprisal Weighting.** Surprisal-based weights emphasize tokens with low conditional probability
 646 under the current policy. These tokens correspond to decisions where small parameter changes can
 647 induce large changes in likelihood, and therefore often dominate gradient variance. Weighting by
 648 surprisal concentrates learning signal on such high-sensitivity decisions.

649 **Entropy-Reduction Weighting.** Entropy-reduction weighting emphasizes tokens that sharply
 650 reduce predictive entropy. These tokens correspond to commitment points in generation, where the
 651 model transitions from a diffuse distribution over continuations to a more concentrated one. This
 652 mirrors the role of branching points in tree-based reasoning methods, but is computed locally from
 653 the model’s predictive distribution without explicit tree construction.

654 C.5 Bias–Variance Tradeoff

655 Both advantage estimation and intrinsic token weighting introduce bias in exchange for variance
 656 reduction. Advantage estimation does so by relying on a learned value function, whose bias can be
 657 substantial when the underlying state abstraction is weak. Intrinsic token weighting introduces bias
 658 by reshaping the contribution of token-level gradients based on heuristic salience measures. The
 659 empirical question is therefore not whether bias is introduced, but whether the bias–variance tradeoff
 660 is favorable.

661 This motivates the empirical comparison implemented in the repository: if intrinsic token weighting
 662 improves accuracy, pass@ k , or optimization behavior under matched settings, then the bias it
 663 introduces may be more benign than the bias induced by ill-defined value targets over text prefixes.
 664 The point is not that weighting is unbiased, but that its inductive bias may be better aligned with
 665 token-level credit assignment in language generation.

666 C.6 Signal Collapse Under Low-Rank Adaptation

667 The preceding analysis assumes that the intrinsic weights $\{w_t\}_{t=1}^T$ are non-degenerate: if every
 668 scheme degenerates to $w_t \equiv 1/T$, then token weighting cannot provide any credit redistribution over
 669 uniform. We now formalize when this degeneracy occurs under LoRA.

670 **Setup.** Let π_{ref} be a frozen base model with last-layer hidden states $h_t^{\text{base}} \in \mathbb{R}^d$ and a LoRA
 671 adapter whose net contribution at position t is $\Delta h_t \in \mathbb{R}^d$, giving adapted hidden states $h_t^{\text{adapted}} =$
 672 $h_t^{\text{base}} + \Delta h_t$ and logits $z_t^{\theta} = W_{\text{lm}} h_t^{\text{adapted}}$. Let $\beta = \max_t \|\Delta h_t\|_2$ denote the maximum per-
 673 position adapter contribution and $L = \|W_{\text{lm}}\|_{\text{op}}$ its logit-space Lipschitz constant. Write $G(\cdot)$ for
 674 the Gini coefficient of a non-negative vector and $\text{EffN}(w)/T = 1/(T \sum_t w_t^2)$ for the normalized
 675 effective-token count.

676 **Proposition 1** (Collapse of surprisal and entropy weighting). *Let $\alpha_t^{\text{surp}} = -\log \pi_{\theta}(y_t \mid y_{<t}, x)$
 677 and $\alpha_t^{\text{surp,ref}} = -\log \pi_{\text{ref}}(y_t \mid y_{<t}, x)$, with corresponding normalized weights w^{surp} and $w^{\text{surp,ref}}$.
 678 Then for every trajectory,*

$$\|w^{\text{surp}} - w^{\text{surp,ref}}\|_1 \leq \frac{4L\beta}{\sum_t \alpha_t^{\text{surp,ref}}}.$$

679 In particular, if the base-model surprisal profile is close to uniform (as it is on content-dense reasoning
680 traces with smooth base-model calibration), $w^{\text{surp}} \rightarrow 1/T$ as $\beta / \min_t \alpha_t^{\text{surp,ref}} \rightarrow 0$. The analogous
681 statement holds for the entropy-reduction score.

682 *Proof sketch.* The softmax Jacobian is 2-Lipschitz in the log-space norm, so the per-token log-
683 probabilities differ by at most $2L\beta$ between the adapted and base model. The projection onto
684 the simplex via within-trajectory normalization is 1-Lipschitz in ℓ_1 up to a factor of $2/\sum_t \alpha_t^{\text{ref}}$;
685 combining these gives the stated bound. $\square$

686 **Proposition 2** (Collapse of divergence weighting). *Let $\alpha_t^{\text{div}} = |\log \pi_\theta(y_t \mid y_{<t}, x) - \log \pi_{\text{ref}}(y_t \mid$
687 $y_{<t}, x)|$ and $w^{\text{div}} = \alpha^{\text{div}} / \sum_t \alpha_t^{\text{div}}$. Then*

$$0 \leq \alpha_t^{\text{div}} \leq 2L\beta \quad \forall t,$$

688 so $\max_t \alpha_t^{\text{div}} / \min_t (\alpha_t^{\text{div}} + \varepsilon) \rightarrow 1$ as $\beta \rightarrow 0$, and consequently $G(w^{\text{div}}) \rightarrow 0$ and $\text{EffN}(w^{\text{div}})/T \rightarrow$
689 1. That is, divergence weighting collapses to the uniform distribution whenever the LoRA adapter’s
690 per-token perturbation norm vanishes uniformly.

691 The key observation is that the intrinsic scores are all determined by the per-token *log-probability*
692 *differential*, which is precisely what LoRA’s rank- r constraint drives small and approximately uniform
693 across positions. The collapse is not a property of a particular weighting scheme; it is a property of
694 measuring per-token variation through the output distribution when the policy has been constrained
695 to a small neighbourhood of the reference.

696 C.7 ARCA and Position-Discriminative Signal

697 ARCA side-steps the collapse by measuring a quantity that is position-varying even when the adapter’s
698 logit-space impact is small.

699 **Proposition 3** (ARCA remains non-degenerate). *Suppose the adapter-induced residual admits*
700 *the decomposition $\Delta h_t = \sum_\ell B_\ell A_\ell x_{\ell,t}^{\text{pre}} + e_t$, where $x_{\ell,t}^{\text{pre}}$ is the pre-adapter activation at layer*
701 *ℓ and position t and $\|e_t\|$ is a higher-order cross-layer term. Let $v_{\ell,t} = B_\ell A_\ell x_{\ell,t}^{\text{pre}}$ and define*
702 *$V_\ell = \text{Var}_t(\|v_{\ell,t}\|)$ and $\bar{V}_\ell = \mathbb{E}_t[\|v_{\ell,t}\|]$. Then*

$$\text{Var}_t(\alpha_t^{\text{ARCA}}) \geq \max_\ell V_\ell - O\left(\sum_\ell \bar{V}_\ell \|e_t\|\right).$$

703 In particular, as long as the pre-adapter activations $x_{\ell,t}^{\text{pre}}$ have non-trivial position-to-position
704 variance at any layer, α_t^{ARCA} has bounded-below variance across positions, and its normalized
705 weights w_t^{ARCA} do not converge to the uniform distribution as the adapter shrinks uniformly.

706 The intuitive content of Proposition 3 is that ARCA inherits its position-discrimination from the
707 base model’s own activation pattern (which is already non-uniform because attention and layer norm
708 routing produce content-versus-filler distinctions) rather than from a quantity that LoRA is specifically
709 constrained to make small. Scaling the adapter weights uniformly scales α^{ARCA} uniformly, so the
710 normalized weights w^{ARCA} are scale-invariant in the adapter magnitude.

711 C.8 Bias–Variance Tradeoff

712 Both advantage estimation and intrinsic token weighting introduce bias in exchange for variance
713 reduction. Advantage estimation does so by relying on a learned value function, whose bias can be
714 substantial when the underlying state abstraction is weak. Intrinsic token weighting introduces bias
715 by reshaping the contribution of token-level gradients based on heuristic salience measures. The
716 empirical question is therefore not whether bias is introduced, but whether the bias–variance tradeoff
717 is favorable—and, per Propositions 1 and 2, whether the bias-inducing signal survives the LoRA
718 bottleneck at all.

719 C.9 Summary

720 This perspective reframes advantage estimation as one particular approach to credit redistribution in
721 policy gradient methods. Intrinsic token weighting offers an alternative that leverages the structure

722 of the language model’s predictive distribution, without assuming the existence of meaningful state
723 values. Under LoRA, however, the output-distribution-based signals that most published weighting
724 methods rely on are driven toward uniform; the adapter-residual signal that ARCA uses is, by
725 Proposition 3, provably non-degenerate in the same regime.

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